

CEMLA Course: Financial Econometrics

PROGRAM

July 22 - 26, 2024 — Digital (Mexico City time, UTC -6)

Course Format:

The presentation slides will be in English but the course will be conducted in Spanish, with simultaneous interpretation. Videoconferences will take place using the Zoom platform. The Course is aimed at analysts, junior researchers, and mid-level officials in economic research, financial stability, risk management, or related areas from CEMLA's central banks membership.

**JULY
22
2024**

09:00-10:30 hrs. SESSION 1. Estimation Methods

Ordinary Least Squares (OLS). Newton-Raphson method. Maximum Likelihood Estimation (MLE). Generalized Method of Moments (GMM).

10:30-11:00 hrs. Break

11:00-12:30 hrs. SESSION 2. Time Series Models

Autoregressive models (AR). Stationarity and seasonality. Vector Autoregression Models (VAR) in reduced and structural form: estimation and identification. Local Projections (LP).

12:30 hrs. Day ends

**JULY
23
2024**

09:00-10:30 hrs. SESSION 3. Expected Returns on Currencies

Expected returns on foreign exchange. Uncovered Interest Parity (UIP). Evidence of the UIP condition. Forward rate. Carry trade.

10:30-11:00 hrs. Break

11:00-12:30 hrs. SESSION 4: Interest Rate Models

Nominal term structure models. Decomposing the zero-coupon interest rate to expected inflation and to risk neutral. Real term structure model.

12:30 hrs. Day ends

**JULY
24
2024**

09:00-10:30 hrs. SESSION 5. Asset Pricing Models

Capital Asset Pricing Model (CAPM). Time varying model. Linear factor model. Multifactor pricing model.

10:30-11:00 hrs. Break

11:00-12:30 hrs. SESSION 6. Volatility Models

Volatility measures. Implied volatility from options prices. ARCH models. GARCH models. Stochastic volatility models.

12:30 hrs. Day ends

**JULY
25
2024**

09:00-10:30 hrs. SESSION 7. Non-Parametric Models

Review of estimation methods (binned means, kernel and series). Estimation of the yield curve. Kernel densities.

10:30-11:00 hrs. Break

11:00-12:30 hrs. SESSION 8. Value at Risk and Quantile Regressions

Value at Risk (VaR) estimation methods. Quantile regression estimation. Application of quantile regression and VaR.

12:30 hrs. Day ends

**JULY
26
2024**

09:00-10:30 hrs. SESSION 9. Non-Linear Models

Threshold Autoregressive Models (TAR): estimation, inference, and testing for non-linearity. Threshold models in panel data.

10:30-11:00 hrs. Break

11:00-12:30 hrs. SESSION 10. Application to Financial Frictions

Models with three regimes, with two threshold variables and dynamic models. Application to investment and financial constraints.

12:30 hrs. Day and course end